
MHF 4U1

CHAPTER 6: EXPONENTIAL AND LOGARITHMIC FUNCTIONS

Day	Section	Homework	Completion
1	6.1 The Exponential Function and its Inverse	Page 320 # 1,2,5-8,10ab,11-16	
1	6.2 Logarithms	Page 328 #1-6,8-10, 13,15	
1	6.3 Transformations of Logarithmic Functions	Page 338 #1,2,4,5,10-13,17,18,20	
1	6.4 Power Law of Logarithms	Page 347 # 1-6e.o., 9-12,15	
1	6.5 Making connections: Logarithmic Scales in the Physical Sciences	Page 353 # 1,2,4,6-12, 22	
1	REVIEW	Page 356 # 1-9,12-18 Page 358 #1-6, 9-11	
1	CHAPTER 6 TEST		

CHAPTER 6 TEST DATE:

6.1 THE EXPONENTIAL FUNCTION AND ITS INVERSE

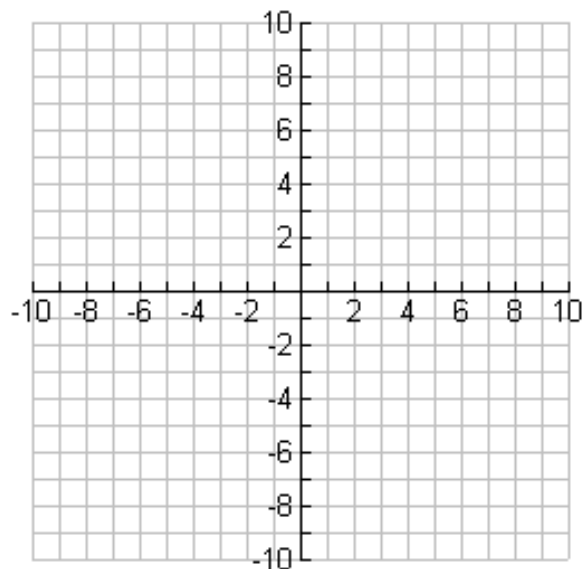
Graph the following exponential functions (in blue) and their inverses (in red) on the grids provided.

Recall: The inverse of a function is a reflection in the line $y = x$.

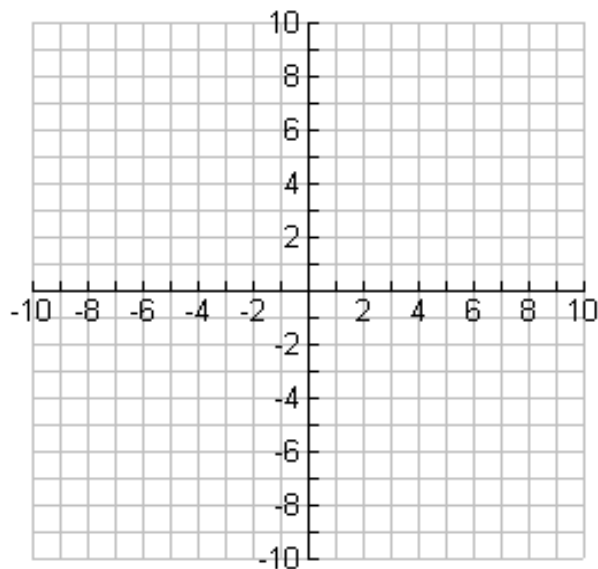
The inverse of a function can be found by **interchanging the x and y values**.

Mapping Rule: $(x,y) \rightarrow (y,x)$

a) $f(x) = 2^x$ and $f^{-1}(x)$



b) $g(x) = \left(\frac{1}{2}\right)^x$ and $g^{-1}(x)$



Properties of the exponential function:

The base "b" is _____

The x-intercept is _____

The y-intercept is _____

The x-axis is _____

If $b > 1$ the function is _____

If $0 < b < 1$, the function is _____

Domain: _____

Range: _____

Exponential equation: $y = b^x$ $\longrightarrow$

Properties of the inverse function:

The base "b" is _____

The x-intercept is _____

The y-intercept is _____

The y-axis is _____

If $b > 1$ the function is _____

If $0 < b < 1$, the function is _____

Domain: _____

Range: _____

Inverse equation: $x = b^y$

The **inverse of the exponential** function is called the _____.

6.2 LOGARITHMS

The Logarithmic Function ($b > 0$ and $b \neq 1$)

in exponential form:

$$x = b^y$$

in logarithmic form:

$$y = \log_b x$$

read as: "the logarithm of x
to the base b"

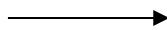
$\log_{\text{base}}$ answer = exponent

The inverse of the exponential function is the logarithmic function. These functions are defined for all bases $b > 0$ and $b \neq 1$. Why are these restrictions in place?

exponential form

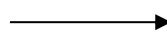
logarithmic form

$$5^3 = 125$$



$$\log_5 125 = 3$$

$$\left(\frac{1}{2}\right)^2 = \frac{1}{4}$$



$$\log_{\frac{1}{2}} \frac{1}{4} = 2$$

Basic properties of logarithms: rewrite in exponential form to solve for x.

$$\log_5 5 = x$$

$$\log_b b =$$

$$\log_8 1 = x$$

$$\log_b 1 =$$

$$\log_3 3^4 = x$$

$$\log_b b^x =$$

Example 1: Evaluate.

a) $\log_2 16$

b) $\log_7 \frac{1}{49}$

c) $\log_{\frac{2}{5}} \frac{8}{125}$

d) $\log_6 1$

e) $\log_3 3^8$

f) $\log_{10} 0.001$

COMMON LOGARITHMS: $\log_{10} x$ written as $\log x$. (The base is understood to be 10.)

Example 2: Use a scientific calculator to evaluate each common logarithm.

g) $\log 0.001$

h) $\log 305$

Example 3: Solving logarithms without a base of 10: rewrite in exponential form then use your calculator as a tool to approximate each solution.

i) $\log_2 10$

j) $\log_3 100$

Try it now!

1. Rewrite $\frac{1}{216} = 6^{-3}$ in logarithmic form. _____

2. Solve the following without a calculator.

$\log \frac{1}{10^3}$	$\log_3 81^2$	$\log_2 \left(\frac{1}{16} \right)^3$	$\log_5 \sqrt{625}$	$\log_{\frac{1}{9}} \left(\sqrt[3]{81} \right)^2$

Try it now!

1. Evaluate without a calculator.

$$\log_4(8\sqrt{2})$$

$$\log_8(4\sqrt{2})$$

2. The function $f(x) = \log(x)$ has the point (10, 1) on its graph. If $f(x)$ is vertically stretched by a factor of 3, reflected in the x-axis, horizontally stretched by a factor of 2, horizontally translated 5 units to the right, and vertically translated 2 units up, determine:

a) equation of the transformed function: _____

b) the coordinates of the image point transformed from (10, 1) _____

c) the domain of the transformed function _____

d) the range of the transformed function _____

6.3 TRANSFORMATIONS OF THE LOGARITHMIC FUNCTION

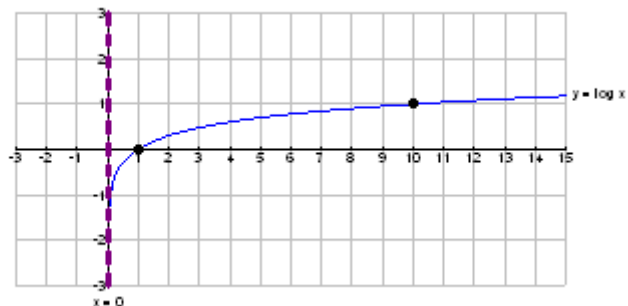
$$y = a \log [k(x - d)] + c$$

mapping rule:

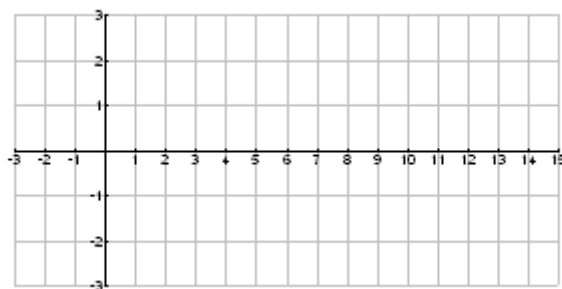
$$(x, y) \Rightarrow \left(\frac{1}{k}x + d, ay + c \right)$$

Given $f(x) = \log x$, sketch the following transformations:

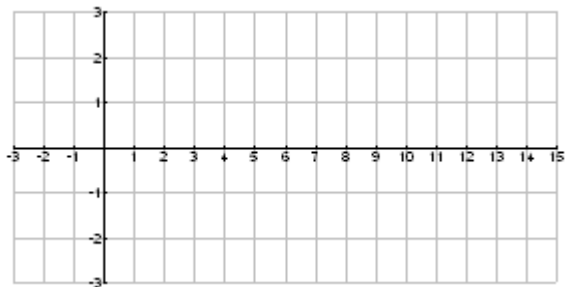
$$y = \log x$$



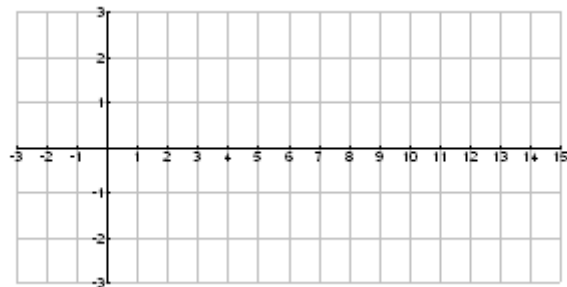
$$y = 2 \log x$$



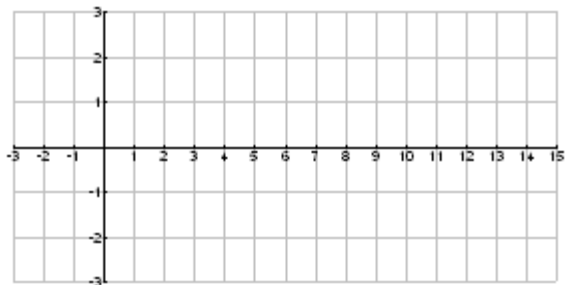
$$y = \log x - 2$$



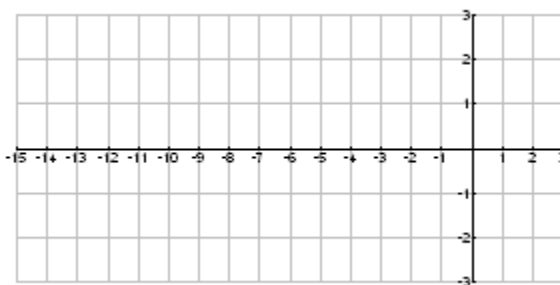
$$y = \log 2x$$



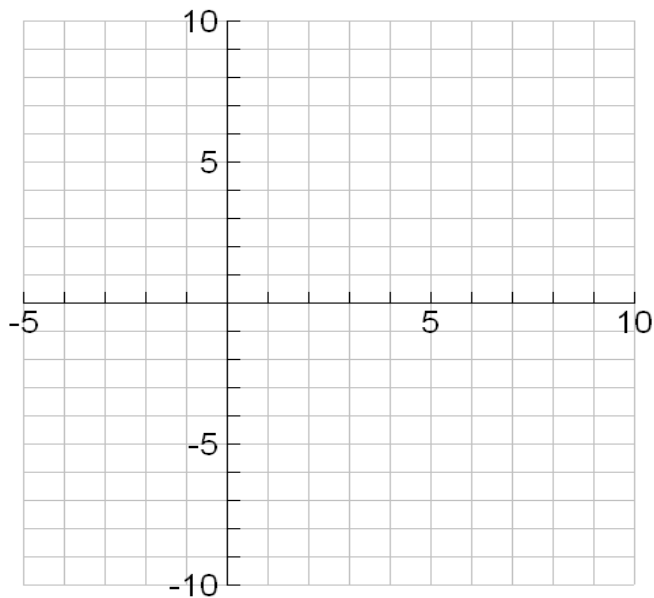
$$y = \log (x - 4)$$



$$y = -\log (-x)$$



Ex. 1: Given the base function $f(x) = \log x$. Sketch the transformation $y = -3 \log(-2x + 12) - 4$. Identify the key features listed below.



Domain: _____

Range: _____

x-intercept: _____

y-intercept: _____

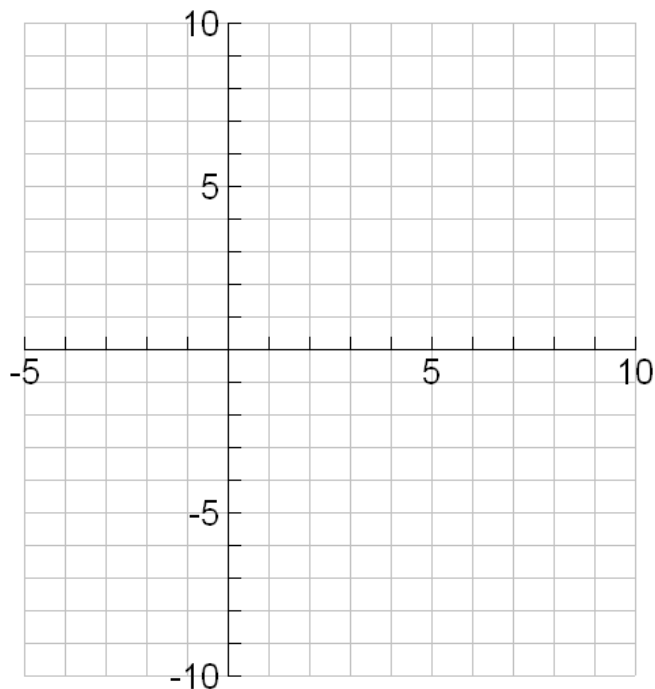
increasing interval: _____

decreasing interval: _____

positive interval: _____

negative interval: _____

Ex. 2: Sketch the transformation $y = \log_3(x - 2) + 1$. Sketch its inverse on the same grid. Find the equation of the inverse function.



Ex. 3: On your own, sketch the transformation $y = 2 \log_{\frac{1}{2}}(x + 1) - 3$.

6.4 LAWS OF LOGARITHMS

The Power Law

In exponential form: $(b^m)^n = b^{mn}$

In logarithmic form: $\log_b m^n = n \log_b m$

Proof:

Example 1: Use the power law to evaluate.

a) $\log_5 25^7$

b) $\log_2 \sqrt[5]{8}$

c) $6^t = 300$

Scientific calculators evaluate only common logarithms (base 10). To evaluate a logarithm with any other base, we must rewrite the problem using a common logarithm.

The Change of Base Formula

$$\log_b m = \frac{\log m}{\log b}, \quad b > 0, b \neq 1, m > 0$$

Proof:

Example 2: Use the change of base formula to evaluate.

a) $\log_2 9$

b) $\log_5 2000$

c) $\log 3^x = 5$

6.5 APPLICATIONS OF LOGARITHMS

Logarithmic scales are used to measure quantities that vary greatly. For example:

Standard numbers:	1	10	100	1000	. . .
Exponential scale:	10^0	10^1	10^2	10^3	. . .
Logarithmic scale:	0	1	2	3	. . .

MAGNITUDE: We use the Richter Scale (0 to 8.9) to measure the magnitude of an earthquake. In general, when comparing two intensities, we are looking for their ratio.

$$M = \log\left(\frac{I}{I_0}\right)$$

compared to the standard

and

$$M_1 - M_2 = \log\left(\frac{I_1}{I_2}\right)$$

comparing 2 intensities

Where **M** is the magnitude and **I** is the intensity.

Example 1: What is the magnitude of an earthquake 1000 times as intense as a standard earthquake?

Example 2: How many more times intense was the 1995 quake in Japan that measured 7.0 on the Richter Scale than the quake in Nevada that measured 5.5?

SOUND: We use the Decibel Scale (dB) to measure the loudness of a sound.

FYI: a measurement of 10 $\Rightarrow$ quiet while a measurement of 200 $\Rightarrow$ extremely loud.

$$L = 10 \log \left(\frac{I}{I_0} \right)$$

compared to the standard

and

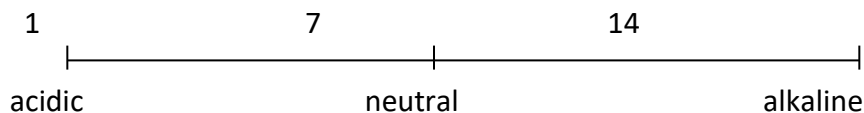
$$L_1 - L_2 = 10 \log \left(\frac{I_1}{I_2} \right)$$

comparing 2 intensities

Where **L** is the sound level in dB and **I** is the intensity of the sound.

Example 3: How much more intense is an ordinary conversation measuring 50 dB than a whisper that measures 20 dB?

pH SCALE: We use the pH scale to measure the acidity or alkalinity of a chemical solution.



$$\text{pH} = -\log [\text{H}^+]$$

Where **H⁺** is the concentration of hydronium ions (measured in moles per litre).

Example 4: Blood has a hydronium ion concentration of 4×10^{-8} mol/L. Is it acidic or alkaline?

Example 5: O.J. has a pH of 3.5. What is the concentration of hydronium ions in O.J.?

Try it now!

1. Express in logarithmic form

(a) $5^2 = 25$ (b) $6^0 = 1$ (c) $49^{\frac{1}{2}} = 7$ (d) $125^{\frac{1}{3}} = 25$

2. Express in exponential form

(a) $\log_3 1 = 0$ (b) $\log_6 \left(\frac{1}{36}\right) = -2$ (c) $\log_9 27 = \frac{3}{2}$

3. Evaluate the following

(a) $\log_7 49$ (b) $\log_2 16$ (c) $\log_3 \sqrt{27}$ (d) $\log_2 \sqrt[3]{32}$ (e) $\log_{17} 1$
 (f) $\log_3 \frac{1}{81}$ (g) $6^{-2\log_6 8}$ (h) $7^{2\log_7 6}$ (i) $\log_3 (9\sqrt{27})$
 (j) $\log_4 (64 \times \sqrt[3]{16})$ (k) $\log_5 \left(\frac{\sqrt[3]{25}}{625}\right)$ (l) $\log_{12} 576 - \log_{12} 4$ (m) $\log_6 18 + \log_6 12$

Answers:

1. (a) $\log_5 25 = 2$ (b) $\log_6 1 = 0$ (c) $\log_{49} 7 = \frac{1}{2}$ (d) $\log_{125} 25 = \frac{2}{3}$
 2. (a) $3^0 = 1$ (b) $6^{-2} = \frac{1}{36}$ (c) $9^{\frac{2}{3}} = 27$
 3. (a) 2 (b) 4 (c) $\frac{3}{2}$ (d) $\frac{5}{3}$ (e) 0 (f) -4
 (g) $\frac{1}{64}$ (h) 36 (i) $\frac{7}{2}$ (j) $\frac{11}{3}$ (k) $\frac{-10}{3}$ (l) 2 (m) 3

Try it now!

TRUE OR FALSE:

- The graph of $f(x) = \log(x-4)$ has a vertical asymptote at $x = -4$.
- If $y = 3^x$, then $x = \log_3 y$.
- If $f^{-1}(x) = 5^x$, then $f(x) = \log 5x$.
- You can write $\log_4 21$ as $\frac{\log 4}{\log 21}$.
- The graph of $f(x) = \log(x+3) - 4$ is the graph of $f(x) = \log x$ translated 3 units left and 4 units down.
- The y-intercept of the exponential function $y = 6^x$ is 1.
- For any values $a > 0$ and $b > 0$, it can be shown that $a = b^{\frac{\log a}{\log b}}$.